

Wild type against a BACH2 knockout: iPSC to T lineage

2 loops compared · [example-ipsc-tcell-wt](#) · [example-ipsc-tcell-bach2](#)

Full decision-by-decision report for each: [example-ipsc-tcell-wt](#) · [example-ipsc-tcell-bach2](#)

Synthetic stand-in reactor. Every number below is the output of a phenomenological stand-in whose per-arm behaviour was invented for software testing. The comparison demonstrates that the loop can find that two genotypes need different conditions. It is not evidence that they do.

What this comparison answers

The first loop asks what a wild-type iPSC-derived T-lineage process reaches on its own, and what the best schedule the search can find for it looks like. The second asks the same of a line carrying a BACH2 knockout that must still be expanded to the same target, alongside its own isogenic control.

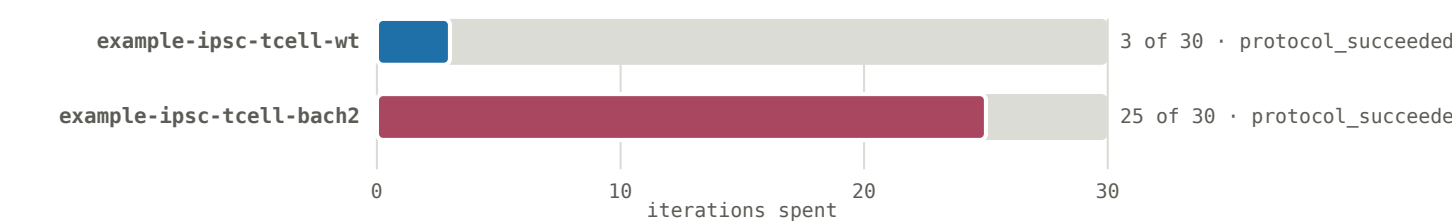
The result worth reading is not either arm's number. It is **whether one shared schedule can serve both genotypes**. Where the search finds that it cannot, it says so explicitly and splits that parameter per arm, and the split is listed below.

What it does not answer

The reactor is a synthetic stand-in whose per-arm behaviour was invented. It was built so that the levers the curated BACH2 annotation suggests are the ones that pay off, which is what makes the demonstration legible and also what makes it worthless as biology. The curated annotations separate what a cited paper reports from a transfer to this cell type, and the cited BACH2 work is in mouse and human peripheral or engineered T cells, not in an iPSC-derived differentiation. No numeric value in either protocol is attributed to any publication, because no full text was retrieved.

Outcomes side by side

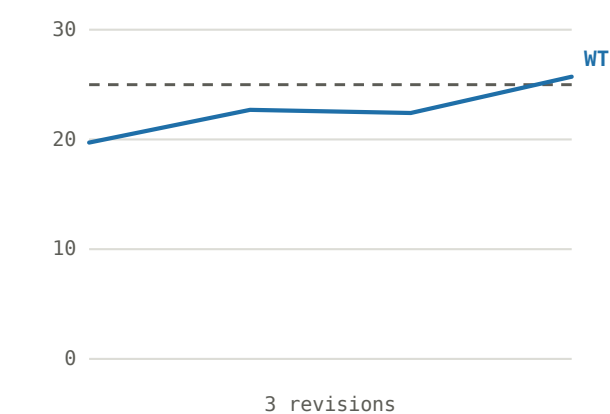
LOOP	VERDICT	ENDED IN	ITERATIONS	BEST PER ARM	DECISIONS
example-ipsc-tcell-wt	SUCCESS	protocol_succeeded	3/30	WT 25.718 (met)	4
example-ipsc-tcell-bach2	SUCCESS	protocol_succeeded	25/30	BACH2_KO 25.954 (met), WT 25.744 (met)	27



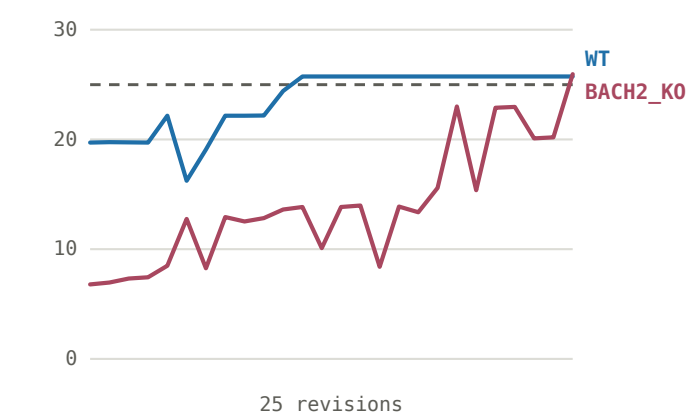
Iterations each loop spent, against the budget its request allowed. The pale bar is the budget. Only a protocol revision spends an iteration; gathering evidence and closing QC do not.

How each search moved

example-ipsc-tcell-wt



example-ipsc-tcell-bach2



The same axis in both panels, so the panels can be compared directly. Separate panels rather than one overlay because the loops ran different arms; a single plot would imply they share a run.

What each search established

example-ipsc-tcell-wt

The search recorded no cross-arm finding.

example-ipsc-tcell-bach2

- IL-7 cannot be set to one shared value: moving it 16 -> 25.6 ng/mL improved BACH2_KO and degraded WT. The arms are being given separate values of this parameter from here on.

Where the arms ended up

The rows below are the per-arm deviations each loop established. A parameter that appears here is one the search could not satisfy with a single shared value across genotypes.

example-ipsc-tcell-wt

No per-arm deviation: one shared schedule served every arm in this loop.

example-ipsc-tcell-bach2 [full report](#)

ARM	PARAMETER	STAGE	SHARED VALUE	THIS ARM	UNIT
BACH2_KO	IL-15	expansion	16	14.08	ng/mL
BACH2_KO	IL-7	expansion	16	25.6	ng/mL
BACH2_KO	anti-CD3	maturation	13.31	11.713	ng/mL
BACH2_KO	IL-7	maturation	10	25.6	ng/mL
BACH2_KO	SCF	maturation	50	80	ng/mL
BACH2_KO	DLL4	t_commitment	500	800	ng/mL
BACH2_KO	IL-7	t_commitment	10	16	ng/mL

"Shared value" is what the base protocol runs for every other arm. A row here is a parameter the search could not satisfy with one value across genotypes.

Evidence standing of each final protocol

LOOP	REPORTED	ADAPTED	DESIGN CHOICE	GAP
example-ipsc-tcell-wt	0	0	29	0
example-ipsc-tcell-bach2	0	0	36	0

example-ipsc-tcell-wt

■ Design choice (29)



29 quantities

Every quantity in the final protocol by where its value came from. A bar that is entirely design choice means nothing in the recipe is attributable to a publication, which is a statement about the evidence available, not about the recipe being wrong.

example-ipsc-tcell-bach2

36 quantities

Every quantity in the final protocol by where its value came from. A bar that is entirely design choice means nothing in the recipe is attributable to a publication, which is a statement about the evidence available, not about the recipe being wrong.

Reported: The value equals a cited claim. **Adapted:** Transferred or converted from cited claims, with a rationale. **Design choice:** Chosen without direct evidence. Needs human approval and carries no citation. **Gap (no value):** No value. Blocks a wet-lab run.

References used across these loops

Publications

1. Fine-tuning BACH2 dosage balances stemness and effector function to enhance antitumor T cell therapy. Nat Immunol. 2025. [10.1038/s41590-025-02389-z](https://doi.org/10.1038/s41590-025-02389-z)
full text not retrieved
 Used for:
 - claim L05 (bach2_dosage_effect)
2. Michaels YS, Edgar JM, Major MC, et al. DLL4 and VCAM1 enhance the emergence of T cell-competent hematopoietic progenitors from human pluripotent stem cells. Sci Adv. 2022;8(34):eabn5522. [10.1126/sciadv.abn5522](https://doi.org/10.1126/sciadv.abn5522) · PMC PMC9401626
full text not retrieved
 Used for:
 - claim L01 (notch_ligand_requirement)
 - claim L02 (stage_order)
3. Roychoudhuri R, Clever D, Li P, et al. BACH2 regulates CD8+ T cell differentiation by controlling access of AP-1 factors to enhancers. Nat Immunol. 2016;17(7):851-860. [10.1038/ni.3441](https://doi.org/10.1038/ni.3441) · PMC PMC4918801
full text not retrieved
 Used for:
 - claim L03 (bach2_terminal_differentiation)

4. Tsukumo S, Unno M, Muto A, et al. Bach2 maintains T cells in a naive state by suppressing effector memory-related genes. Proc Natl Acad Sci USA. 2013;110(26):10735-10740. [10.1073/pnas.1306691110](https://doi.org/10.1073/pnas.1306691110) · PMC PMC3696756
full text not retrieved

Used for:

- claim L04 (bach2_naive_maintenance)

Annotation sources

Each annotation carries its own confidence. `local_annotation` restates what a cited paper reports; `inference` is a transfer to this cell type or readout that no cited paper measures; `synthetic_fixture` is invented for software testing.

1. Roychoudhuri et al. 2013, Nature 498:506-510, doi:10.1038/nature12199 (PMID 23728300) - reference key roychoudhuri2013 [tcell_curated_genes]
 - BACH2 knockout: increase target_marker_pct [local_annotation]
2. Roychoudhuri et al. 2016, Nat Immunol 17:851-860, doi:10.1038/ni.3441 (PMID 27158840) - reference key roychoudhuri2016 [tcell_curated_genes]
 - BACH2 knockout: increase growth_rate_per_h [local_annotation]
3. Transfer by the curator from roychoudhuri2016 (doi:10.1038/ni.3441) and bach2_dosage_2025 (doi:10.1038/s41590-025-02389-z) to an iPSC-derived expansion readout. No cited paper measures this. [tcell_curated_genes]
 - BACH2 knockout: decrease target_cells_per_input_cell [inference]
4. Tsukumo et al. 2013, PNAS 110:10735-10740, doi:10.1073/pnas.1306691110 (PMID 23754397) - reference key tsukumo2013 [tcell_curated_genes]
 - BACH2 knockout: decrease target_marker_pct [local_annotation]

Limitations of the comparison

- The arms were compared inside a synthetic stand-in whose per-arm knobs were invented. The stand-in was built so that the levers the annotation suggests are the ones that pay off, which is what makes the demonstration legible and also what makes it worthless as biology.
- One replicate per arm. Every difference is directional only.

- The curated gene annotations distinguish what a cited paper reports from a transfer to this cell type. Only the former rests on a publication, and the cited work is in peripheral or engineered T cells, not in an iPSC-derived differentiation.
- A parameter the search split per arm is a statement about this stand-in's response surface, not about the genotype.

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